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BRIEF REPORT

# Defining Worst-Case-Scenario Thresholds in Soccer: Intensity Versus Volume

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***Purpose:*** This study aimed to enhance the understanding of soccer match peak demands by describing worst-case scenario (WCS) and time spent above 80% and 90% of the WCS for total distance (TD) and high-speed running (HSR). The investigation considered playing level (first team vs under-19 [U19] team) and playing position (center backs, fullbacks, midfielders, and forwards) to assess how WCS and the time spent above specific thresholds vary across different populations. ***Methods:*** Data from 31 players in a professional Italian soccer club were collected during the 2022–23 season. Microtechnology devices tracked physical activity during matches. Players were categorized by position, and WCS was determined using rolling averages over a 1-minute period. Time spent above 80% and 90% of WCS for TD and HSR was calculated. ***Results:*** The U19 team exhibited higher HSR WCS compared with the first team (~63 m·min$^{-1}$ vs ~56 m·min$^{-1}$). Midfielders recorded the highest TD WCS (~208 m·min$^{-1}$), and forwards exhibited the highest HSR WCS (~70 m·min$^{-1}$). The first team spent significantly more time above 80% (~6 min) and 90% (~1 min) of TD WCS. Midfielders spent significantly more time above the 80% (~7 min) of TD WCS, while forwards above the 80% (~2 min) of HSR WCS. ***Conclusions:*** The study emphasizes that WCS used alone may not sufficiently capture real match intensity. Considering the time spent above specific thresholds provides additional insights (ie, between-levels differences and position). Practitioners should consider both WCS and time spent above thresholds for individualized training prescriptions, reflecting differences in playing roles.

***Keywords:*** match, peak demand, Global Positioning System

Soccer training and match running demands are typically analyzed and compared based on total or high-speed running (HSR) distances, normalized per minute (in meters per minute), to enable easier comparisons.[1] The limit of this method is that it underestimates the real peak demands.[2] Indeed, the stochastic nature of a soccer match leads to fluctuations of intensity which players need to be prepared for such intermittent activities.

Recognizing the limitations of activity per minute, the concept of worst-case scenario (WCS) has been introduced to analyze the peak running demands over a specific time window.[3,4] Previous research has established soccer matches WCS using rolling averages for different time windows such as 1, 3, or 5 minutes. Novak et al[5] reported differences between playing positions and in relation to other contextual factors as the phases of the match.

However, as highlighted by Johnston et al,[6] the "peak" collected through the WCS only occurs once throughout the match, neglecting the overall demand of a soccer match, specifically for the volume of activities completed in proximity to this "peak." Moreover, this approach showed high variability and lack of association with internal responses.[5]

Up to now, the volume-related component of the peak intensity has not received specific attention in football. Indeed, understanding how much time the players spend above a specific intensity relative to the peak period would offer greater insight to training prescription. While the approach of time spent in different intensity zone is now commonly used for internal load (ie, time spent above 90% of maximal heart rate vs absolute heart rate values), little is done in that direction regarding external load. Only Johnston et al[6] and Couderc et al,[7] respectively, in rugby league and rugby Seven investigated this approach to overcome the limitation of WCS. However, Johnston et al[6] investigated the distribution of distance accumulated in relation to the maximal mean 1-minute period, while Couderc et al[7] presented their findings in in meters·per minute.

In contrast, this study aims to introduce a novel method for quantifying the time spent in proximity of the WCS. The objective is to estimate the duration (in minutes) spent above 80% and 90% of the WCS during official matches, according to the playing level and playing position in elite football players.

## Methods

### Subjects

Data were collected during the 2022/23 soccer season (July 2022–May 2023) from 31 playing in the first team (n: 18, age: 24.2 [4.2] y, body mass: 79.7 [5.5] kg, height: 184.5 [5.2] cm) and U19 team (n: 13, age: 18.1 [0.7] y, body mass: 76.7 [9.2] kg, height: 182.9 [7.6] cm) of a professional Italian soccer club. Data arose from daily monitoring throughout the season. Therefore, ethics committee clearance was not required.[8] However, all data were anonymized, and the study was conducted according to the Declaration of Helsinki.

### Methodology

Microtechnology devices with a sampling frequency of 10 Hz (WIMU PRO; RealTrack Systems) were used to collect the

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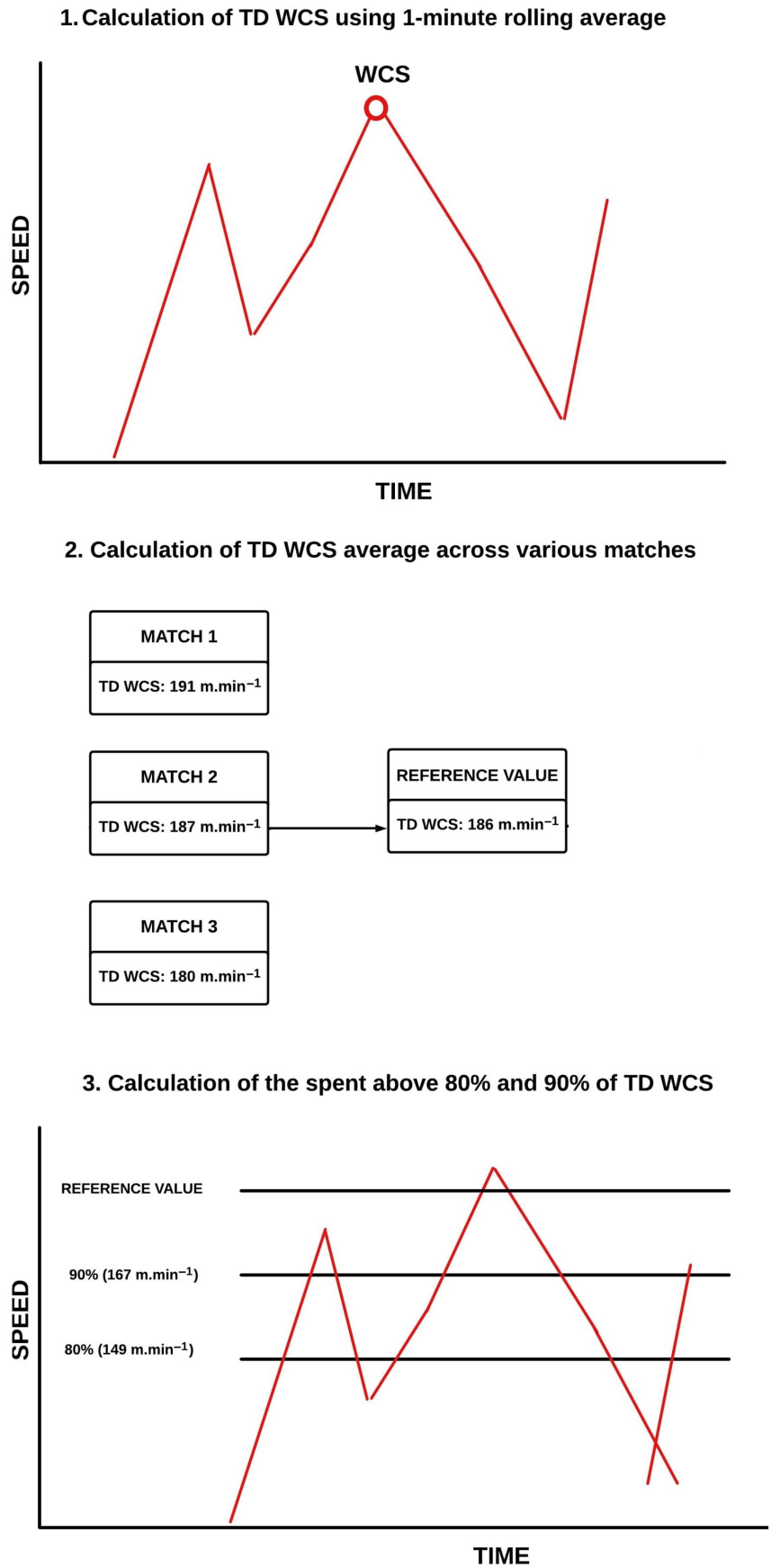


**Figure 1** — Summary of the steps performed to calculate the time spent above the 80% and 90% of WCS. TD indicates total distance; WCS, worst-case scenario.


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physical activity of players during the matches (first team: n = 44, U19: n = 40). Only matches where players completed the entire match were considered in the following study. This approach was used to prevent the inclusion of substitute players who might demonstrate greater physical capabilities than the starters, potentially due to pacing strategies.[9] A total of 337 (10 [9] per player) observations were recorded. Players were categorized according to playing position: center-backs (n = 7), full-backs (n = 6), midfielders (n = 8), and forwards (n = 10).

After each match, locomotor variables were calculated using raw data. The locomotor variables of interest were total distance (TD) and HSR distance defined as the distance covered above 19.8 km·h$^{-1}$. A 1-minute rolling average was then employed, with the maximum value used to determine the WCS, representing the maximum distance covered in TD and HSR within the 1-minute time window.[4] The average of the WCS recorded across various matches was calculated. This reference value was used to determine the time spent above 2 distinct thresholds: 80% and 90% of WCS. The duration during which players recorded physical activity above these thresholds was computed and expressed in minutes. All analyses were performed using the Anaconda (version 3.9.12, Anaconda Inc) and Python (version 3.9.19, Python Language Reference, available at http://www.python.org) libraries. All steps previously performed are summarized in Figure 1.

## Statistical Analysis

A within-subject linear mixed model was used to analyze mean differences of TD and HSR WCS, and time spent above 80% and 90% of WCS with respective confidence intervals of 95% in relation to playing level (first team vs U19 team) and playing position (center backs vs fullbacks vs midfielders vs forwards). Players' identity was inserted into the model as random effect to consider the repeated measurements. When statistically significant differences were found, the least significant difference approach to multiple comparisons was used.[10] Standardized effect sizes (ESs) were also calculated, and values of .2, .5, and .8 were interpreted as small, moderate, and large differences, respectively.[11] SPSS software was used to perform the statistical analyses. The threshold for statistical significance was set at $P < .05$.

## Results

Table 1 displays the 1-minute TD and HSR WCS values, along with the corresponding time spent above 80% and 90% of TD and HSR WCS.

Statistical analysis revealed no significant differences in TD WCS between the first team and the U19 team. However, HSR WCS was significantly higher in the U19 team ($P < .01$, ES = 0.88). Analyzing WCS differences based on playing position, midfielders exhibited the highest TD WCS ($P < .01$, ES = 5.92), while forwards showed the highest HSR WCS ($P < .01$, ES = 1.94).

Further exploration of time spent above WCS thresholds indicated that the first team spent significantly more time above 80% of TD WCS ($P < .05$, ES = 0.27). Additionally, the first team spent more time above 90% of TD WCS ($P < .01$, ES = 0.29). Examining playing positions, midfielders spent more time above 80% of TD WCS ($P < .01$, ES = 0.57), while no differences were found above 90% of TD WCS. Conversely, forwards experienced more time above 80% of HSR WCS ($P < .01$, ES = 0.58). Supplementary Table S1 (available online) presents all pairwise comparisons related to WCS values and the time spent above WCS thresholds.

**Table 1 Descriptive Statistics (Mean [SD]) of WCS Values and Time Spent Above the Different WCS Thresholds for the 1-Minute Time Window**

| Playing level | Variable | Mean (SD) |
|---|---|---|
| First team | TD WCS, m·min$^{-1}$ | 192.22 (12.24) |
| | HSR TD, m·min$^{-1}$ | 56.28 (4.63) |
| | TD WCS, 80%; min | 6.40 (3.45) |
| | HSR WCS, 80%; min | 1.31 (1.05) |
| | TD WCS, 90%; min | 1.46 (1.18) |
| | HSR WCS, 90%; min | 0.80 (0.81) |
| U19 team | TD WCS, m·min$^{-1}$ | 193.87 (11.35) |
| | HSR TD, m·min$^{-1}$ | 63.62 (10.78) |
| | TD WCS, 80%; min | 5.56 (2.55) |
| | HSR WCS, 80%; min | 1.37 (1.11) |
| | TD WCS, 90%; min | 1.16 (0.85) |
| | HSR WCS, 90%; min | 0.77 (0.77) |
| **Playing position** | **Variable** | **Mean (SD)** |
| Center backs | TD WCS, m·min$^{-1}$ | 180.41 (3.92) |
| | HSR TD, m·min$^{-1}$ | 52.16 (5.31) |
| | TD WCS, 80%; min | 5.45 (2.66) |
| | HSR WCS, 80%; min | 1.15 (0.97) |
| | TD WCS, 90%; min | 1.20 (0.94) |
| | HSR WCS, 90%; min | 0.71 (0.68) |
| Lateral backs | TD WCS, m·min$^{-1}$ | 190.96 (3.79) |
| | HSR TD, m·min$^{-1}$ | 60.11 (3.87) |
| | TD WCS, 80%; min | 5.70 (3.16) |
| | HSR WCS, 80%; min | 1.24 (0.96) |
| | TD WCS, 90%; min | 1.30 (1.06) |
| | HSR WCS, 90%; min | 0.73 (0.79) |
| Midfielders | TD WCS, m·min$^{-1}$ | 208.23 (5.36) |
| | HSR TD, m·min$^{-1}$ | 62.52 (8.89) |
| | TD WCS, 80%; min | 7.20 (3.42) |
| | HSR WCS, 80%; min | 1.40 (1.17) |
| | TD WCS, 90%; min | 1.45 (1.10) |
| | HSR WCS, 90%; min | 0.84 (0.90) |
| Forwards | TD WCS, m·min$^{-1}$ | 196.28 (5.15) |
| | HSR TD, m·min$^{-1}$ | 70.62 (5.68) |
| | TD WCS, 80%; min | 5.56 (2.66) |
| | HSR WCS, 80%; min | 1.78 (1.19) |
| | TD WCS, 90%; min | 1.36 (1.15) |
| | HSR WCS, 90%; min | 0.94 (0.77) |

Abbreviations: HSR, high-speed running; TD, total distance; WCS, worst-case scenario.

## Discussion

The current study aimed at: (1) quantifying WCS for TD and HSR based on playing level and playing position and (2) evaluating the time spent above 80% and 90% of WCS. Notably, the study not only identified differences in WCS but also revealed distinctions


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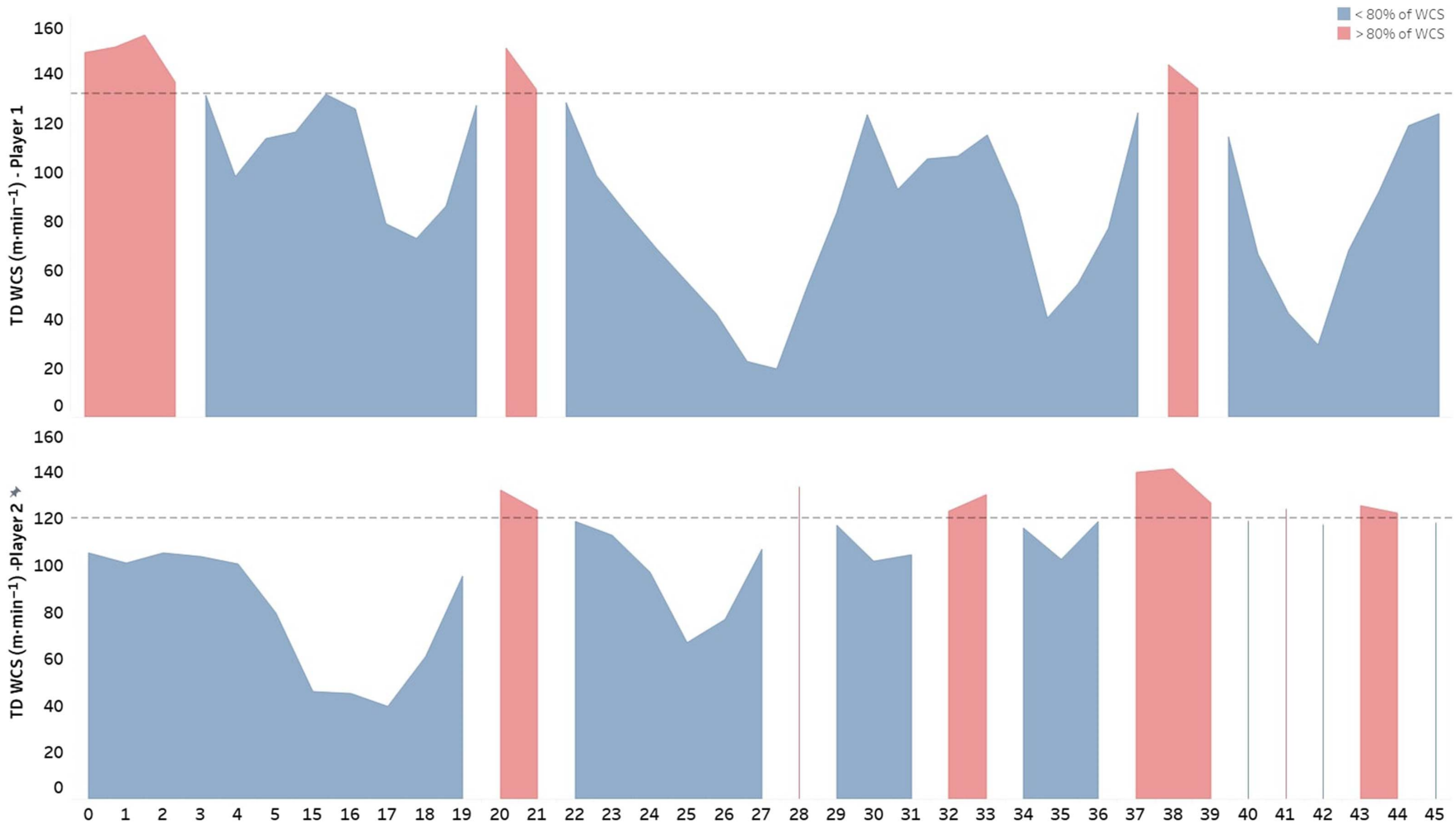


**Figure 2** — Example of 2 different players showing similar TD WCS (~150 m·min$^{-1}$) during the first half of 1 match but different time spent above 80% of the WCS (player 1 [~7 min] vs player 2 [~11 min]). Periods where the player reaches 80% of the WCS are marked in red, while periods below this threshold are depicted in blue. See article online for the color version of the figure. TD indicates total distance; WCS, worst-case scenario.

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in the time spent above 80% and 90% of WCS, highlighting the relevance of monitoring time spent in close to maximal WCS in elite football players.

Previous research has widely analyzed the WCS during soccer matches, and in line with previous findings,[12,13] differences were observed based on playing level and position. Interestingly, in the current study, no significant differences were identified for TD WCS according to playing level, while U19 team exhibited a higher HSR WCS compared with the first team. Regarding the playing position, in agreement with previous results,[14] midfielders traveled the longest average distance during matches (TD WCS), followed by forwards. Conversely, in terms of HSR, forwards recorded the highest WCS, followed by midfielders and full-backs.[15]

While WCS is valuable in characterizing the most intense period of a match, it solely represents a singular peak, providing limited information about the volume of work performed close to these peak match demands. In alignment with Johnston et al,[6] this study aimed to extend the concept of WCS by incorporating information about the volume spent above 80% and 90% of WCS, offering valuable insights for sport practitioners for training prescription. However, Johnston et al[6] applied a distinct methodology, analyzing the distribution of relative intensity among rugby and Australian football athletes. Instead, our study introduces a novel approach by estimating the minutes spent in proximity to the WCS.

Although no differences were found for TD WCS between the first team and the U19 team, the first team spent significantly more time above 80% and 90%, highlighting the need to prepare first team players for repeating peak demands on a more frequent basis. Regarding playing position, noteworthy findings include only midfielders spending significantly more time above 80% of TD WCS, while forwards recorded the highest time spent above 80% of HSR WCS. Interestingly, no differences emerged at the 90% threshold. To build on the insights from this study, future research should explore how the duration spent exceeding various thresholds correlates with physiological and internal load responses.

## Practical Applications

When prescribing training sessions, not only WCS but also the volume of activities performed close to WCS should be considered by soccer practitioners. Indeed, as shown in Figure 2, 2 players can have similar WCS (and overall activity), but different time spent above 80% of the WCS, probably leading to very different internal load and fatigue. This nuanced understanding of time distribution in different thresholds could allow for individualized training sessions according to playing roles, closely replicating real match efforts, intensity, and density. Moreover, this approach can be adapted using different WCS time window calculations (eg, 3, 5, and 10 min).

## Conclusions

Although the worst-case scenario (WCS) might not capture the complete intensity and density of real matches, our study introduces a novel approach to bridge this gap by quantifying the time spent beyond 80% and 90% of WCS thresholds. This analysis revealed differences not only in WCS but also in the time spent across the different thresholds, influenced by players' levels and positions.

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